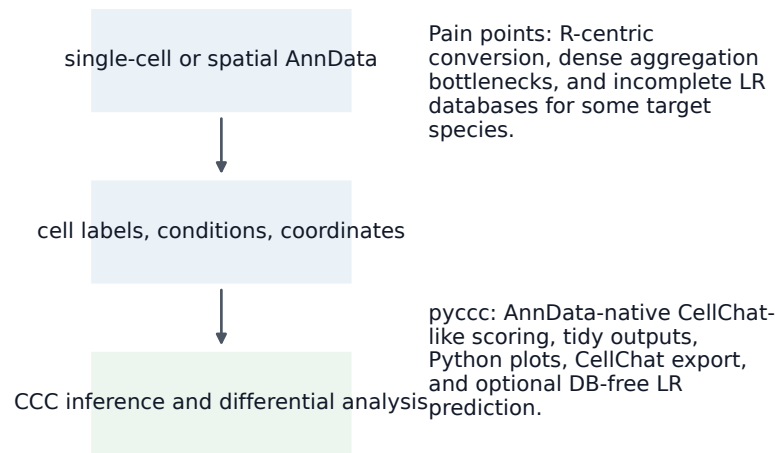
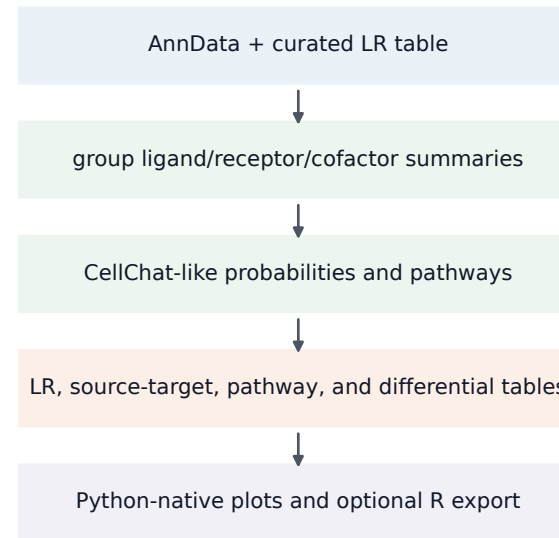


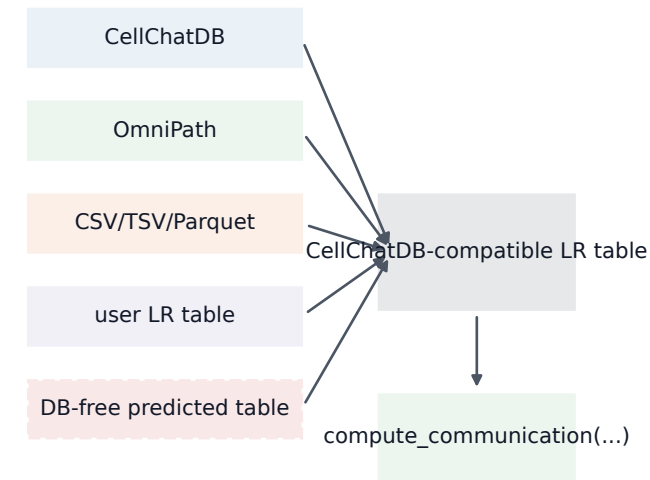
A Problem setting



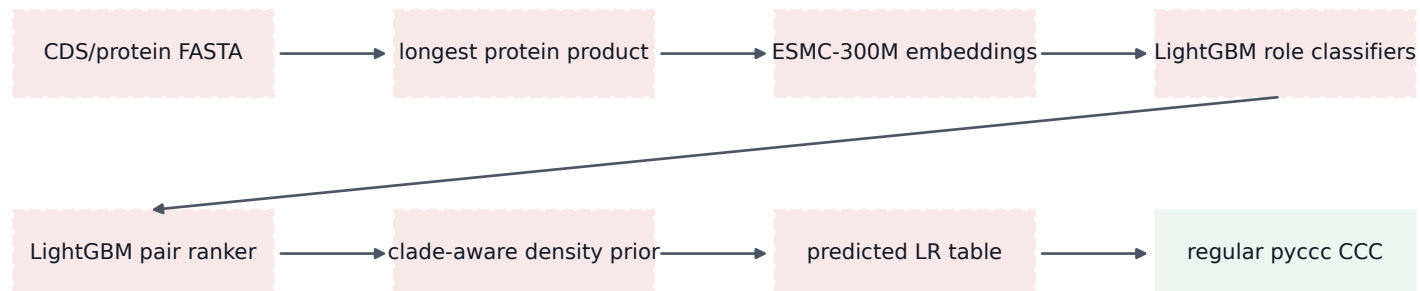
B Regular-mode pipeline



C LR resource layer



D Optional target-species DB-free branch



Dashed boxes indicate the optional experimental LR-table generator. Regular-mode benchmarks use curated LR tables, not DB-free prediction.

E Minimal API

```
import pyccc as pc

db = pc.load_cellchatdb("human")
res = pc.compute_communication(
    adata, groupby="cell_type", lr_table=db)
diff = pc.compare_samples(
    adata_a, adata_b, groupby="cell_type", lr_table=db)

predicted_db = pc.predict_lr_dbfree(
    adata, protein_fasta="target.fa")
report = pc.validate_spatial_lr_table(
    adata, lr_table=predicted_db)
```